

## HW 3

Team members: Catherine Chen, Dylan Pan, Hsiao-Chi Chen, Jackie Zhang, Zara Nip

### Macro factors

#### 1. Asset universe

The dataset contains 12 monthly FRED series from January 2010 to December 2024, with January 2025 to January 2026 used for out-of-sample testing. It covers global equities, rates, FX, and commodities. Equity exposure is proxied by OECD share-price indices for eight major markets; rates exposure is represented by a synthetic 10-year Treasury total return constructed from constant-maturity yields. OECD equity indices are used because the public FRED historical data for BAML credit total returns and the S&P 500 were truncated before 2017 at the time of collection.

**Table: Asset Universe**

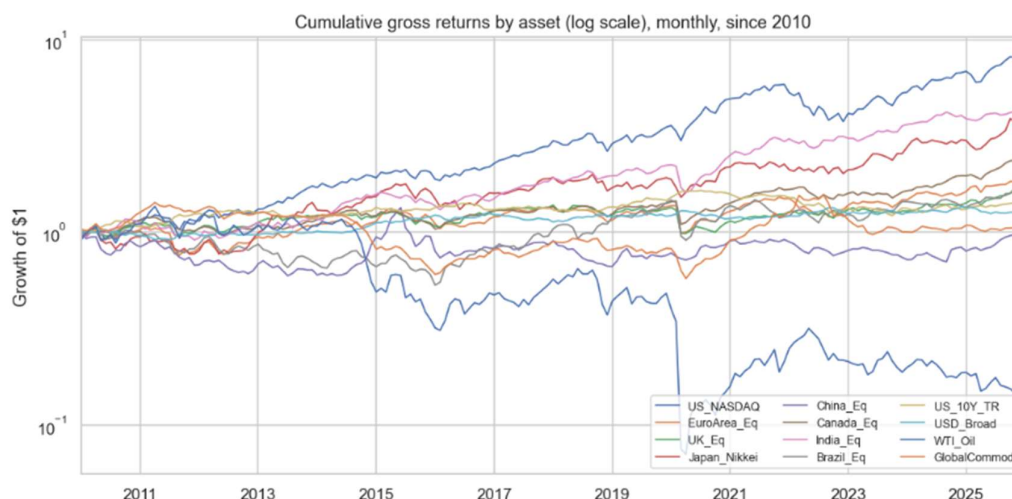
| Variable name | Full name                                    | Source / series                     | Asset class | Region          |
|---------------|----------------------------------------------|-------------------------------------|-------------|-----------------|
| US_NASDAQ     | NASDAQ Composite Index                       | FRED: NASDAQCOM                     | Equity      | United States   |
| EuroArea_Eq   | Euro Area share prices (OECD)                | FRED: SPASTT01EZM661N               | Equity      | Euro Area       |
| UK_Eq         | UK share prices (OECD)                       | FRED: SPASTT01GBM661N               | Equity      | United Kingdom  |
| Japan_Nikkei  | Japan Nikkei share prices (OECD)             | FRED: SPASTT01JPM661N               | Equity      | Japan           |
| China_Eq      | China share prices (OECD)                    | FRED: SPASTT01CNM661N               | Equity      | China           |
| Canada_Eq     | Canada share prices (OECD)                   | FRED: SPASTT01CAM661N               | Equity      | Canada          |
| India_Eq      | India share prices (OECD)                    | FRED: SPASTT01INM661N               | Equity      | India           |
| Brazil_Eq     | Brazil share prices (OECD)                   | FRED: SPASTT01BRM661N               | Equity      | Brazil          |
| US_10Y_TR     | US 10-year Treasury total return (synthetic) | FRED: DGS10 (duration-approximated) | Rates       | United States   |
| USD_Broad     | Broad US Dollar Index                        | FRED: DTWEXBGS                      | FX          | Global (USD)    |
| WTI_Oil       | WTI Crude Oil spot price                     | FRED: DCOILWTICO                    | Commodities | Global (Energy) |
| GlobalCommod  | Global All-Commodities Price Index           | FRED: PALLFNINDEXM                  | Commodities | Global (Broad)  |

#### 2. Data processing

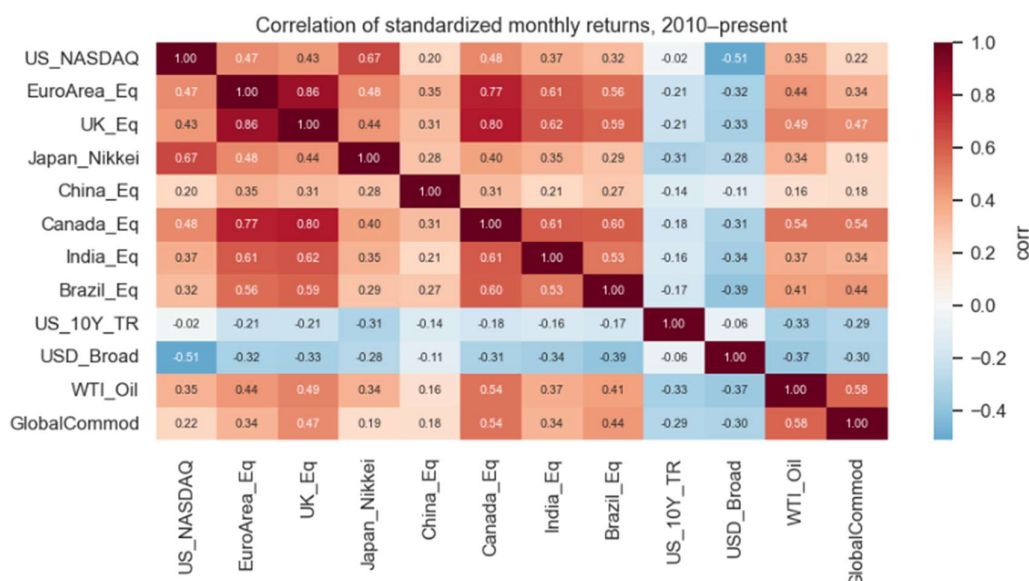
Monthly returns are computed from end-of-month prices, with log changes used for price-based series. We scale each return series by its full-sample standard deviation

to make regression coefficients and PCA loadings comparable across assets. This is particularly necessary given that realized monthly volatility ranges from 1.6% for the broad USD index and 2.0% for the 10-year Treasury to 11.3% for WTI crude.

The log-scale cumulative-return chart shows substantial cross-sectional dispersion since 2010. US NASDAQ dominates cumulative performance, while Brazil and China equities rank among the weakest. Major stress episodes, including the 2014–16 oil collapse and the 2020 COVID drawdown, are clearly visible. In contrast, the broad USD index and 10-year Treasury total return show relatively muted cumulative drift.



The correlation matrix motivates the factor analysis. Equities form a clear positive block, with pairwise correlations mostly between 0.4 and 0.7. The broad USD index is negatively correlated with most non-US equities, around  $-0.3$  to  $-0.5$ , reflecting the translation effect of a stronger dollar on foreign-currency returns. The 10-year Treasury is mildly negatively correlated with risk assets, while WTI crude is modestly positively correlated with equities and the global commodity index.



Overall, the matrix points to a small set of macro blocks: risk-on equities, defensive rates/USD exposure, and the commodity cycle. We formalize this structure using factor regressions and PCA.

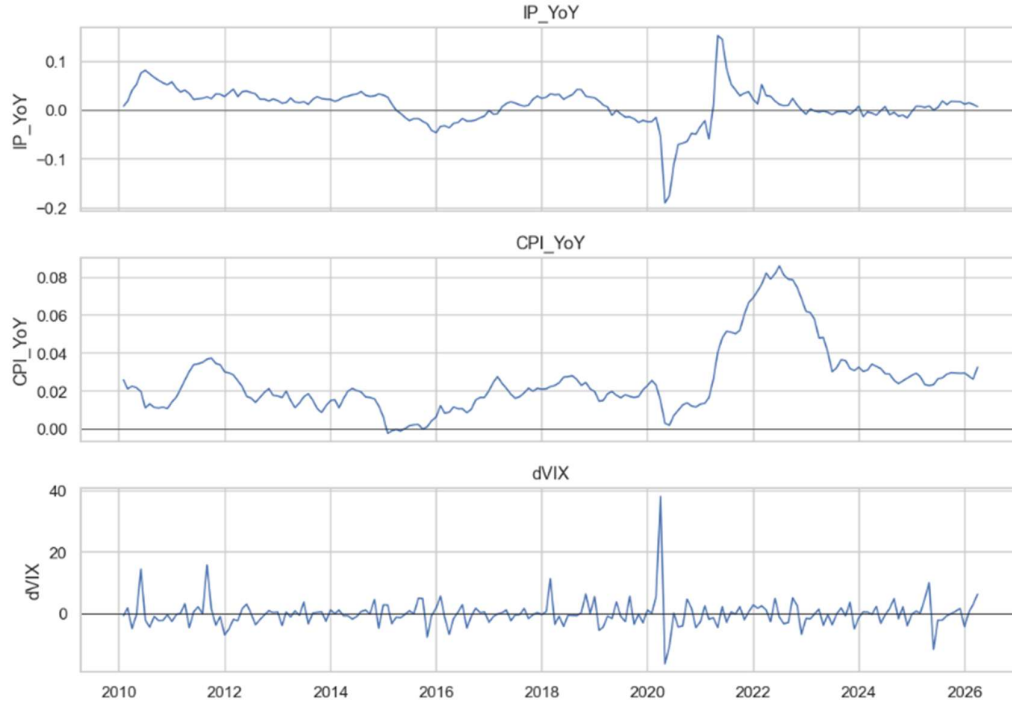
### 3. Macro factors

We use three monthly FRED series to capture growth, inflation, and risk conditions: Industrial production (INDPRO), Headline CPI (CPIAUCSL) and VIX (VIXCLS). INDPRO is transformed into 12-month log growth and serves as the growth factor. CPIAUCSL is transformed into 12-month log inflation and serves as the inflation factor. VIXCLS is averaged monthly and first differentiated, so  $\Delta VIX$  captures changes in financial stress.

The expected signs are economically intuitive. Growth should load positively on equities and commodities. Inflation should load negatively on nominal bonds and positively on commodities.  $\Delta VIX$  should load negatively on risk assets and positively on defensive assets such as the USD and Treasuries.

The plots support these expectations. IP YoY captures the 2020 contraction and 2021 rebound. CPI YoY captures the post-COVID inflation surge, peaking near 9% in mid-2022 before declining.  $\Delta VIX$  is mean-reverting and spikes during major risk-off episodes, including the euro-area crisis, the 2015–16 oil sell-off, the 2018 Q4 drawdown, the March 2020 COVID shock, and the 2022 rate-hike volatility. The transformed factors appear stationary and suitable for OLS regression.

**Chart: History Data for Industrial production, CPI, and VIX**



#### 4. Regression analysis

For each asset, we estimate the time-series regression

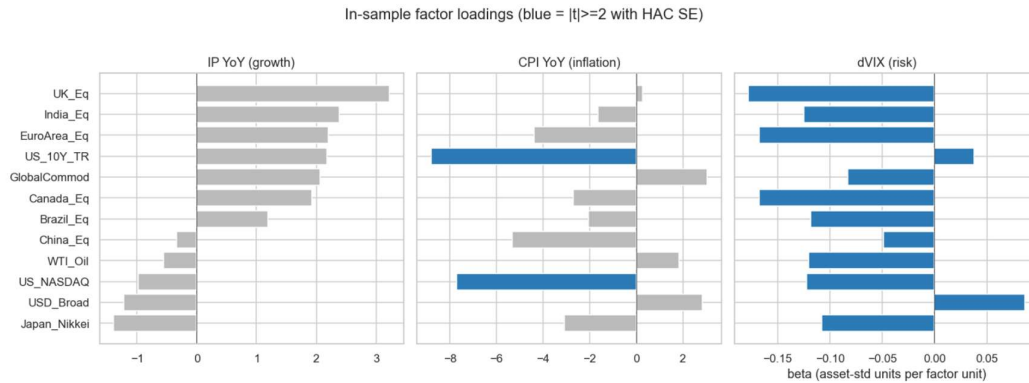
$$\tilde{r}_{i,t} = \alpha_i + \beta_i^{\text{IP}} \Delta \text{IP}_t + \beta_i^{\text{CPI}} \Delta \text{CPI}_t + \beta_i^{\text{VIX}} \Delta \text{VIX}_t + \varepsilon_{i,t}$$

where  $\tilde{r}_{i,t}$  is the standardized return in units of the asset's own monthly standard deviation. Because the dependent variable is standardized, all coefficients are directly comparable across assets: a  $\beta=0.3$  means a one-unit change in the factor moves the asset by 0.3 of its monthly standard deviation. We report six-lag Newey–West t-statistics to account for persistence in monthly macro variables.

The loadings broadly match the economic priors. Equities generally load positively on IP growth and negatively on  $\Delta \text{VIX}$ . The 10-year Treasury loads negatively on inflation, with HAC  $|t| = 2.0$ , while the broad USD loads positively on  $\Delta \text{VIX}$ , consistent with safe-haven behavior.  $\Delta \text{VIX}$  is the strongest factor statistically: ten of twelve assets have significant loadings, with all equity loadings negative and the USD loading positive. Growth and inflation are less consistently significant, which is expected at a noisy monthly horizon.

**Table: Regression Summary**

|              | b_IP   | t_IP   | b_CPI  | t_CPI  | b_dVIX | t_dVIX  | R <sup>2</sup> |
|--------------|--------|--------|--------|--------|--------|---------|----------------|
| asset        |        |        |        |        |        |         |                |
| US_NASDAQ    | -0.982 | -0.661 | -7.710 | -2.114 | -0.122 | -5.232  | 0.344          |
| EuroArea_Eq  | 2.205  | 1.308  | -4.392 | -1.300 | -0.167 | -11.935 | 0.575          |
| UK_Eq        | 3.229  | 1.820  | 0.282  | 0.117  | -0.178 | -17.387 | 0.636          |
| Japan_Nikkei | -1.400 | -0.919 | -3.080 | -1.207 | -0.108 | -4.984  | 0.264          |
| China_Eq     | -0.344 | -0.207 | -5.316 | -1.138 | -0.049 | -2.185  | 0.060          |
| Canada_Eq    | 1.923  | 1.180  | -2.688 | -0.753 | -0.168 | -12.890 | 0.578          |
| India_Eq     | 2.390  | 0.820  | -1.633 | -0.574 | -0.125 | -10.712 | 0.308          |
| Brazil_Eq    | 1.187  | 0.575  | -2.046 | -0.848 | -0.118 | -7.256  | 0.278          |
| US_10Y_TR    | 2.182  | 1.042  | -8.817 | -2.022 | 0.038  | 2.364   | 0.055          |
| USD_Broad    | -1.220 | -0.771 | 2.825  | 0.713  | 0.086  | 4.942   | 0.155          |
| WTI_Oil      | -0.559 | -0.270 | 1.849  | 0.418  | -0.121 | -4.584  | 0.293          |
| GlobalCommod | 2.068  | 0.776  | 3.056  | 0.384  | -0.083 | -7.704  | 0.141          |



Model fit differs sharply across assets. UK equity has the highest  $R^2$  at 0.64, followed by Canada at 0.58 and the Euro Area at 0.57, consistent with their strong exposure to the global risk cycle captured by  $\Delta VIX$ . The weakest fits are the US 10-year Treasury total return and China equity, both with  $R^2=0.06$ , suggesting that their monthly movements are mostly driven by factors outside this three-factor model.

Overall, the macro factors explain about 6–64% of monthly variation across assets. The remaining residual variation motivates the PCA decomposition below.

## 5. Sample period discussion

We use the full January 2010–December 2024 window for in-sample estimation. Although this period spans different regimes, including the zero-lower-bound era and

the 2020–2022 COVID-inflation shock, shortening the sample to 2015–2024 would remove about one-third of the observations. It also changes most coefficients by less than one HAC standard error. The gain in regime homogeneity is therefore smaller than the loss of statistical power.

For this reason, we keep the full sample and treat 2020–2022 as a structural break to interpret with caution, rather than excluding it.

## **6. Out-of-sample forecast**

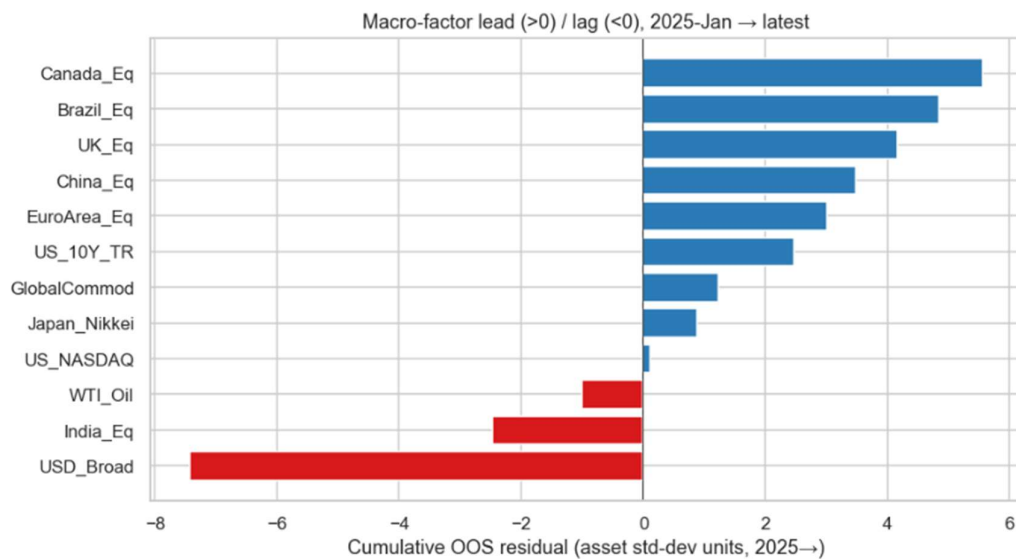
Using the in-sample estimated coefficients, we compute fitted returns for each asset in each OOS month. The OOS residual captures the return not explained by the macro model. We then cumulate residuals from January 2025 through January 2026 to measure each asset’s lead or lag in standard-deviation units. Positive cumulative residuals indicate outperformance relative to the macro-implied return; negative residuals indicate underperformance.

The largest leaders are Canada equity ( $+5.56\sigma$ ), Brazil equity ( $+4.84\sigma$ ), UK equity ( $+4.16\sigma$ ), and China equity ( $+3.48\sigma$ ). The largest laggards are the broad USD index ( $-7.43\sigma$ ) and India equity ( $-2.46\sigma$ ). US NASDAQ is essentially neutral ( $+0.12\sigma$ ), with realized returns close to the model-implied path.

Because the OOS window contains only 12 months’ data, these results should be read as directional lead/lag signals rather than precise statistical estimates.

### **Chart: Cumulative OOS Performance vs. Macro-Factor Prediction**

|              | cum_actual_std | cum_pred_std | cum_resid_std |
|--------------|----------------|--------------|---------------|
| Canada_Eq    | 7.46           | 1.90         | 5.56          |
| Brazil_Eq    | 5.76           | 0.92         | 4.84          |
| UK_Eq        | 5.41           | 1.25         | 4.16          |
| China_Eq     | 3.50           | 0.02         | 3.48          |
| EuroArea_Eq  | 4.24           | 1.23         | 3.01          |
| US_10Y_TR    | 3.28           | 0.81         | 2.47          |
| GlobalCommod | 1.83           | 0.59         | 1.24          |
| Japan_Nikkei | 2.74           | 1.86         | 0.88          |
| US_NASDAQ    | 3.03           | 2.91         | 0.12          |
| WTI_Oil      | -0.83          | 0.17         | -1.00         |
| India_Eq     | 0.48           | 2.94         | -2.46         |
| USD_Broad    | -6.20          | 1.23         | -7.43         |



## 7. Economic interpretation

The main OOS pattern appears to be a US-centered drawdown driven by tariff and trade-policy uncertainty. This shock is largely outside our three-factor macro model, which includes IP growth, CPI inflation, and  $\Delta VIX$  but no direct trade-policy channel. The model therefore over-predicts assets exposed to US policy risk and under-predicts assets that benefited from rotation away from US markets.

This explains why Canada, Brazil, UK, and China equities show large positive residuals: they outperformed their macro-implied returns as capital rotated away

from US assets. India equity is the main exception, likely reflecting domestic-specific headwinds and its high predicted return, with  $\text{cum\_pred\_std} = 2.94\sigma$ , the highest in the universe. The broad USD index lagged most sharply at  $-7.43\sigma$ , consistent with a US asset sell-off in which policy uncertainty weakened the dollar despite weaker risk sentiment. This reversal of the dollar's usual safe-haven behavior lies outside the model's in-sample conditioning.

## **8. Trade implications**

If we interpret the OOS residuals as transitory mispricing, the natural contrarian trade is to short the largest leaders and long the largest laggards. This implies shorting Canada and Brazil equities against long positions in the broad USD index and India equity, with positions sized to be dollar-volatility-neutral.

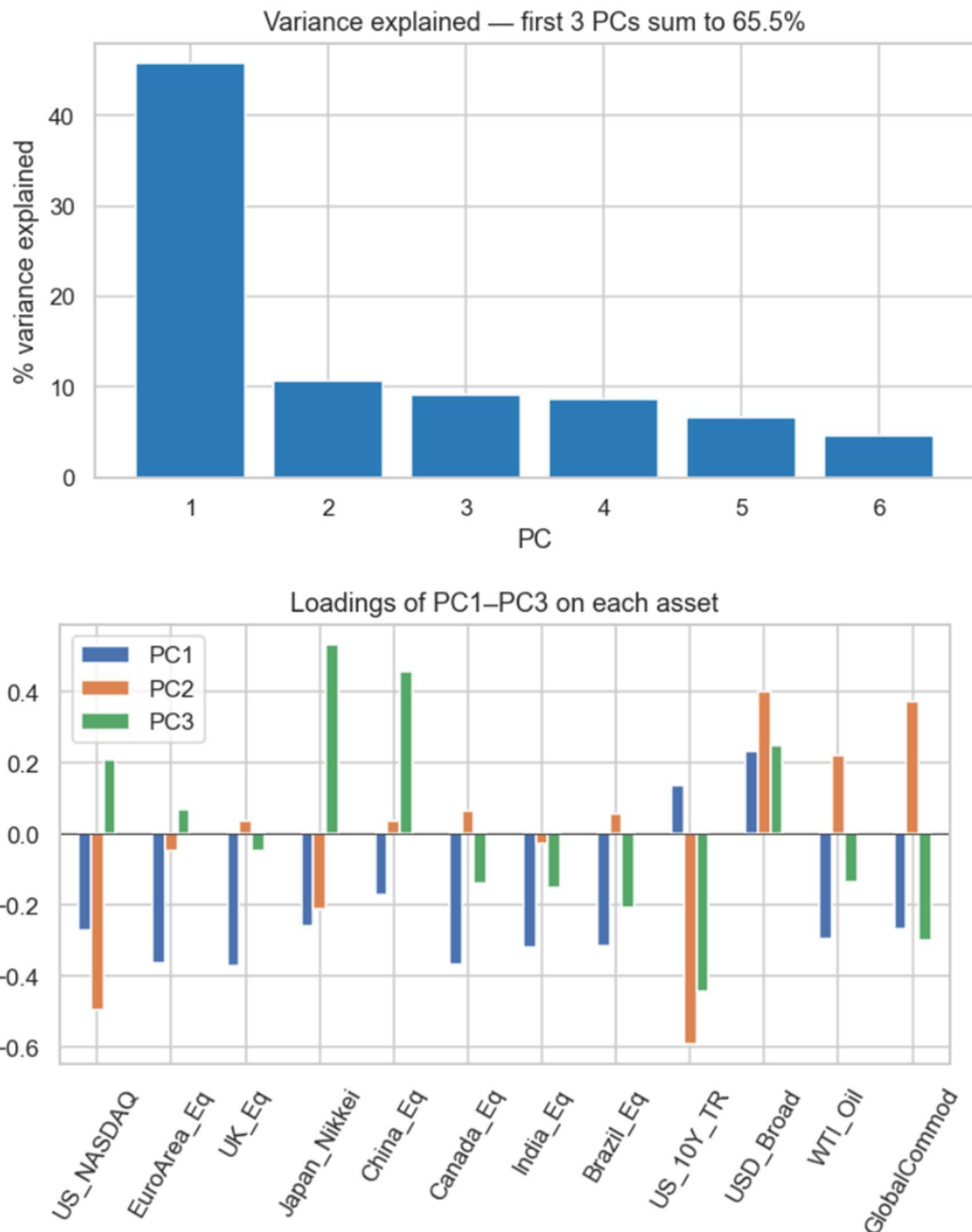
We treat this as a candidate trade, not a recommendation. Residual mean reversion is weak in monthly macro-factor models, especially over a 3–6 month horizon. More importantly, the residuals likely reflect an omitted structural factor — trade-policy uncertainty — rather than noise. If that shock persists, the USD may continue to lag and non-US equity leaders may continue to outperform.

The trade should therefore not be implemented without further validation. The next step is to test whether the residuals proxy for omitted trade-policy risk, ideally by adding a trade-policy uncertainty index to the model.

## **PC price factors**

### **9-11. PC analysis on the same estimation sample data**

We use the same asset universe and run PCA on the standardized in-sample return panel from January 2010 to December 2024. Since each asset is scaled to unit standard deviation, this is equivalent to PCA on the correlation matrix. The first six PCs explain 45.74%, 10.70%, 9.05%, 8.64%, 6.67%, and 4.68% of monthly return variation. The first three PCs explain 65.5% in total, indicating that most cross-asset co-movement is captured by a small number of latent factors.



PC1 is the dominant global risk factor. It loads in the same direction on most equities and commodities, and in the opposite direction on the broad USD index and the 10-year Treasury. Since PCA signs are arbitrary, this can be read as a standard risk-on/risk-off factor: risk assets move together, while defensive USD and Treasury exposure moves against them.

PC2 captures a USD/commodity strength factor. It loads positively on the broad USD index, WTI oil, and the global commodity index, and negatively on US NASDAQ,

Japan, and the 10-year Treasury. This suggests a cycle in which dollar and commodity strength coincides with weaker duration and growth-equity exposure.

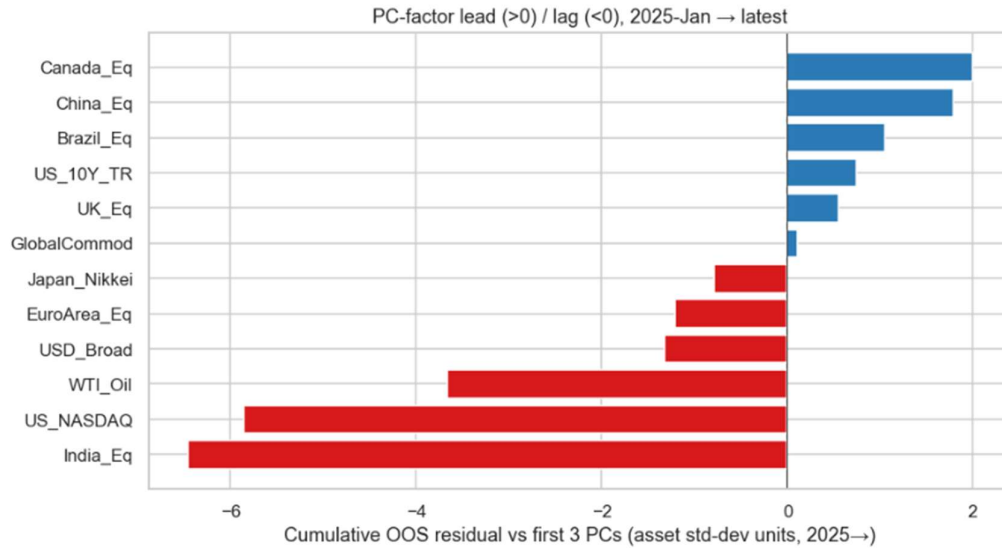
PC3 captures regional and commodity-linked rotation. It loads positively on Japan, China, US NASDAQ, and the broad USD index, but negatively on Brazil, India, Canada, WTI oil, and the global commodity index. This component reflects rotation across equity regions and commodity-sensitive assets.

## 12. PC-Implied Asset Performance and Leading/Lagging Analysis

We treat the first three PC scores as latent cross-asset factors and regress each asset's standardized in-sample return on these scores. Using the in-sample PCA structure, we then compute OOS fitted returns for 2025 onward. The OOS residual captures the portion of each asset's return not explained by the three PCs.

**Chart: Cumulative OOS Performance vs. PC-Factor Prediction**

|              | cum_actual_std | cum_PCpred_std | cum_resid_std | in_sample_R2 |
|--------------|----------------|----------------|---------------|--------------|
| Canada_Eq    | 8.41           | 6.41           | 2.00          | 0.77         |
| China_Eq     | 3.90           | 2.11           | 1.79          | 0.39         |
| Brazil_Eq    | 5.85           | 4.79           | 1.06          | 0.58         |
| US_10Y_TR    | 3.66           | 2.92           | 0.74          | 0.75         |
| UK_Eq        | 6.35           | 5.80           | 0.55          | 0.76         |
| GlobalCommod | 1.88           | 1.77           | 0.11          | 0.65         |
| Japan_Nikkei | 5.84           | 6.63           | -0.79         | 0.77         |
| EuroArea_Eq  | 5.13           | 6.33           | -1.21         | 0.74         |
| USD_Broad    | -5.58          | -4.25          | -1.33         | 0.58         |
| WTI_Oil      | -1.03          | 2.63           | -3.66         | 0.55         |
| US_NASDAQ    | 3.96           | 9.82           | -5.86         | 0.77         |
| India_Eq     | 1.05           | 7.51           | -6.46         | 0.57         |



Positive cumulative residuals indicate cross-asset leaders; negative residuals indicate laggards. The clearest leaders are Canada equity ( $+2.00\sigma$ ) and China equity ( $+1.79\sigma$ ), with Brazil equity, the US 10-year Treasury, and UK equity also modestly positive. The clearest laggards are India equity ( $-6.46\sigma$ ) and US NASDAQ ( $-5.86\sigma$ ), followed by WTI oil ( $-3.66\sigma$ ) and the broad USD index ( $-1.33\sigma$ ).

The PC model fits the in-sample data well, with a median  $R^2$  of about 0.69. Its OOS residuals are therefore smaller and more concentrated than the macro-model residuals, highlighting assets whose 2025+ returns diverged most from the main cross-asset factors.

### 13. Market Interpretation of Asset Deviations

The PCs were estimated from the 2010–2024 covariance structure, so they mainly capture the pre-2025 cross-asset co-movement pattern. However, the 2025 OOS period appears to be affected by a trade-policy shock and regional rotation that are not fully represented by the historical PCs. As a result, several assets moved away from their historical PC-implied behavior, creating clear lead/lag residuals.

On the leading side, Canada equity, China equity, and Brazil equity show large positive residuals. This is consistent with capital rotating away from US-centered policy risk and into non-US regional equity markets. Canada and China are especially strong leaders, suggesting that their realized returns outperformed what the historical PC structure would have predicted. US 10Y Treasury total return also shows a positive residual, which may reflect a flight-to-quality component as investors reduced exposure to riskier US equity assets.

On the lagging side, India equity and US NASDAQ show the most negative residuals. India is the largest laggard, likely because its realized performance was weaker than its PC-implied return and may have been affected by domestic-specific headwinds. NASDAQ also lagged sharply, consistent with the idea that US growth equities were particularly exposed to tariff and trade-policy uncertainty. WTI oil and the broad USD index also lag, suggesting that the 2025 shock was not a standard risk-off episode: instead of the dollar benefiting from safe-haven demand, policy uncertainty weighed on US assets and weakened the usual USD safe-haven channel.

Overall, the lead/lag pattern points to a structural break in cross-asset correlations. The historical PCs still capture the main pre-2025 covariance structure, but they do not fully capture the regional equity rotation and US-centered policy shock that dominated the OOS period.

#### **14. Trading Implications from PC Factor Deviations**

The PC residuals suggest a possible contrarian trade: long the laggards and short the leaders, sized to be dollar-volatility-neutral. We would implement this only as a small tactical overlay, given the substantial idiosyncratic noise left unexplained by the PC model.

Our preferred version uses a double-confirmation filter: long assets that lag both the macro and PC models, and short assets that lead both. This reduces reliance on model-specific residuals. Overall, the trade should remain modest, since post-2024 dispersion appears to be driven by trade-policy uncertainty that neither model captures directly.